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# Advanced CFD Homework #3

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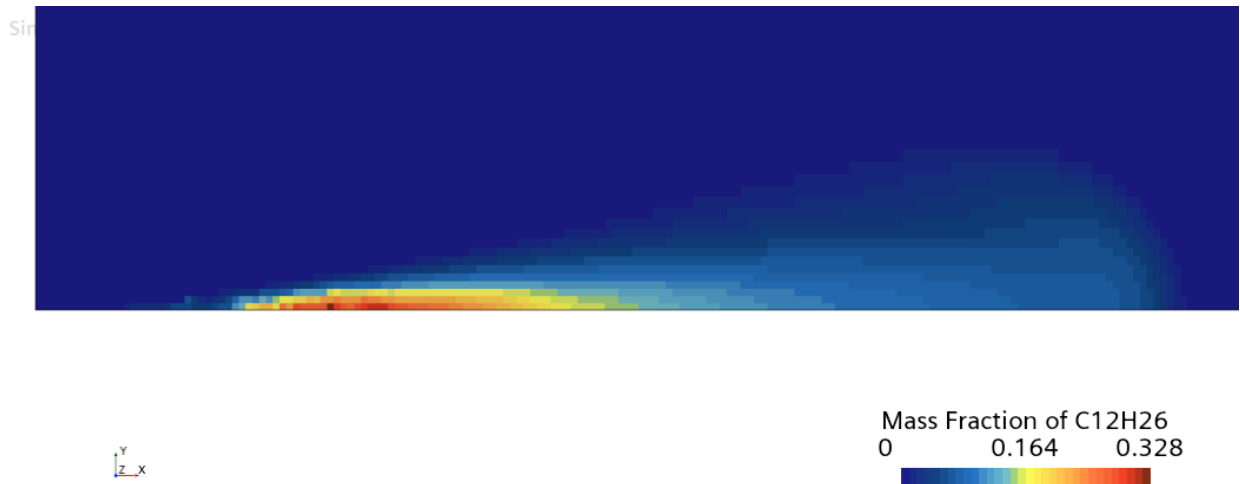


Figure 1: Scalar scene showing Mass fraction of  $C_{12}H_{26}$  particles at 1 [ms]

## 1 Introduction

The objective of this study is to replicate the Sandia Spray A experiment [1] by simulating the spray behaviour and combustion in a wedge constant volume chamber. Part 1 focuses on implementing the Lagrangian spray model to simulate the injection and dispersion of fuel particles into the chamber. In Part 2 the combustion reaction of the fuel spray is analyzed. Consider [2] and [3] showcasing the simulation procedure.

## 2 Part 1 - Lagrangian Spray Modeling

Modelling fuel injection and spray behaviour is important for combustion systems, such as diesel engines. The following section describes how lagrangian spray modelling is implemented to replicate the Sandia Spray A experiment [1] where n-dodecane fuel spray interacts within a non-reacting combustion environment. Consider Figure 1 showing the resulting mass fraction of  $C_{12}H_{26}$  after using Lagrangian spray modelling.

Lagrangian spray modelling consists of sub-models used to treat fuel droplets as discrete individual particles so that it can model the complex behaviour of the spray such as droplet breakup, evaporation, and turbulent dispersion. These sub-models consist of the material particles model, Quasi-Steady Evaporation model, Reitz-Diwakar breakup model, Turbulent dispersion model and more. The predictions of vapour and liquid penetration lengths of these models are then compared against experimental data to address their accuracy.

### 2.1 Simulation Results

Figure 2 shows the liquid and vapour penetration lengths, and reveals a strong correlation between the CFD simulation and the experimental data. One indicator of this strong

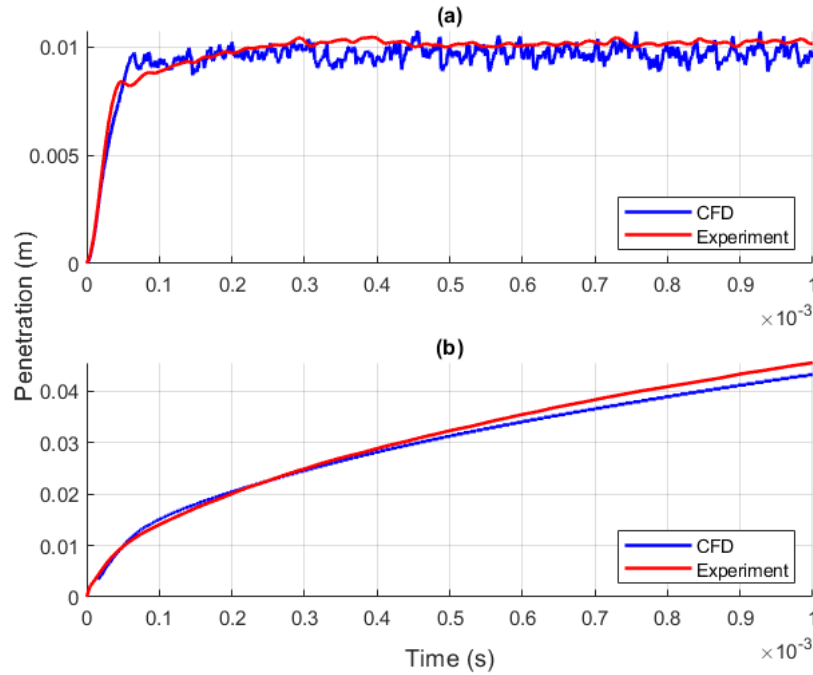


Figure 2: Comparison of CFD result with experimental data from [1] for (a) liquid penetration length and (b) vapour penetration length.

overlap is the Pearson correlation coefficient yielding a value of 0.800 for the liquid penetration and 0.999 for vapor penetration. Such values indicate that little to no discrepancies are present in the vapour penetration, while liquid penetration may be suggested to inaccuracies.

### 2.1.1 Possible discrepancies

One source of error could be in the choice of the Reitz-Diwakar spray breakup model. It models specifically the bag breakup and the stripping breakup meaning it has a direct impact on the liquid-phase dynamics. If the model predicts too much fragmentation, the initial droplet size distribution will evaporate faster leading to a smaller penetration length. On the other hand, too little fragmentation would increase the penetration length. This model goes hand in hand with the quasi-steady evaporation model as it dictates the rate at which particles evaporate and therefore also directly impacts the penetration length. A rapid evaporation rate would cause the droplets to transition from their liquid to vapour phase quickly. This is a less likely source of error seeing that the correlation between the vapor penetration lengths has a strong correlation. Another source of discrepancy could be the turbulent dispersion model and injector setting which dictate the spread of the spray. Overall, the discrepancy for both penetration lengths resembles the experiment results well and gives minuscule discrepancies.

### 2.1.2 Design Engineer Perspective

A diesel engine design engineer looks to increase the efficiency of the engine by establishing optimal conditions for efficient combustion. Efficient combustion is when the energy from the fuel is efficiently converted into useful work without forming unwanted byproducts such as soot or other unburned hydrocarbons [4]. This is known as complete combustion.

The liquid and vapour penetration lengths influence the combustion by determining how efficiently the fuel is mixed with the air, and hence how it combusts. The optimal penetration lengths are ones where the fuel mixes completely with the air before reaching the farthest parts of the chamber. Fuel droplets must evaporate completely before the ignition to allow for the fuel vapour to mix with the air. Therefore, the engineer is interested in the rate of evaporation. Figure 2 indicates that roughly between  $0.1 - 0.2 \times 10^{-3}$  [s] complete evaporation is achieved because liquid penetration stops progressing as the fuel is entirely vaporized. Using this information the engineer can then establish the time for ignition such that minimal byproducts are formed and fuel is not wasted.

If the liquid penetration length is longer, a slower evaporation rate is implied. The prolonged presence of liquid fuel causes soot formation, and overall less efficient combustion. On the other hand, a longer vapour penetration length implies a faster evaporation length and hence more optimal combustion.

## 2.2 Sensitivity Analysis

The simulation has been conducted with changes to three parameters to establish their influence on the result. Firstly, the constant  $C1e$  in the standard K-Epsilon model has been reverred to its original value of 1.44 instead of 1.52 recommended by [2]. Based on the results shown in Figure 3, it can be seen that both the liquid and vapour penetration are underestimated by the default value of 1.44. The value recommended by [2] shows a significantly stronger correlation with the experimental results.

In addition, the simulation was repeated with an increased time step from  $1 \times 10^{-6}$  to  $2 \times 10^{-6}$  [s]. The larger time step shows an overshoot in the liquid penetration plot at approximately  $0.8 \times 10^{-3}$  [s], which is not present in the smaller time step. This overprediction of the result is even more clear when looking at the vapour penetration, which shows a deviation of roughly 0.01 [m]. For this reason, the time step will not be increased for further adjustments to the simulation.

Lastly, the mesh has been stretched further by increasing the spacing from 0.25 [mm] to 0.125 [mm]. This change however has shown no difference in the results, which is why the data is seen to overlap perfectly in Figure 3. This implies that the solution is independent of the mesh and therefore can be computed with the larger spacing of 0.25 [mm] to yield the same result.

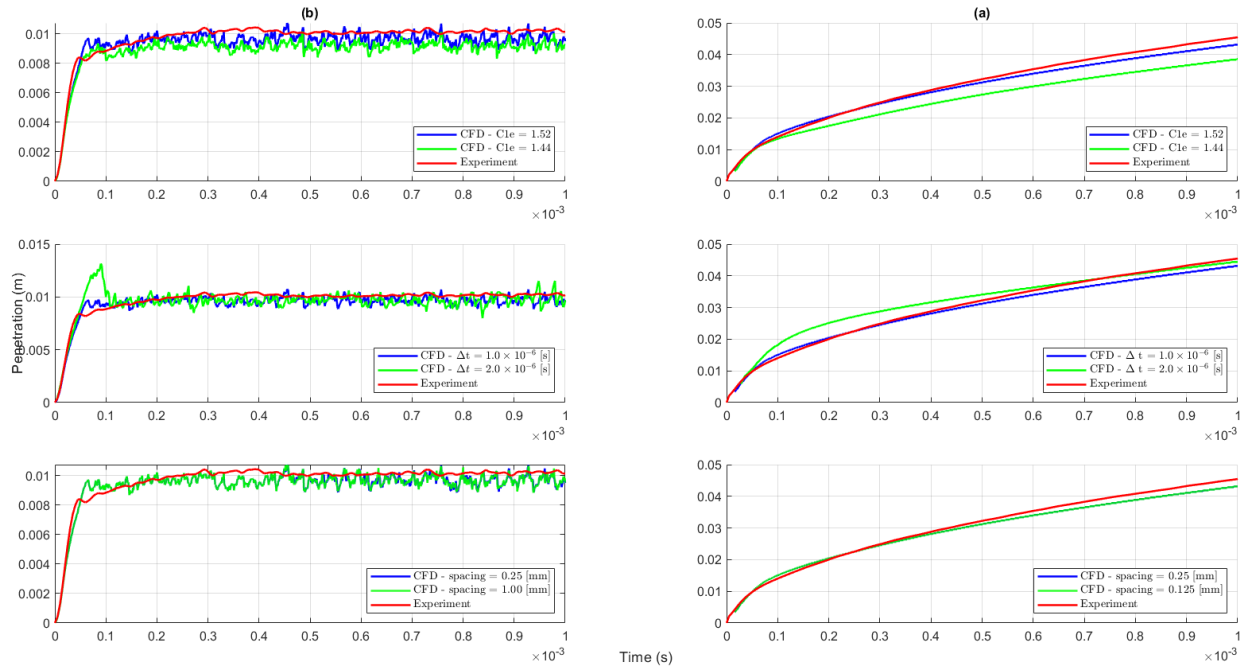


Figure 3: Simulation results showing (a) liquid penetration length in column 1 and (b) vapour penetration length in column 2 for the following adjustments; changing  $k - \epsilon$  solver constant  $C1\epsilon$ , time step, mesh stretching.

## 2.3 Resulting droplet sizes

Figure 4 shows the particle size distribution in a histogram form; where the frequency of the particle's diameter is shown for a time of 1 [ms]. The results indicate that the most likely particle diameter is approximately  $2.6 \times 10^{-6}$ . As the simulation evolves the liquid droplets are more likely to collide and break up into smaller particle sizes, which is why smaller particle sizes are more probable. Smaller particles are preferred because they have a larger surface area and hence increase the evaporation rate.

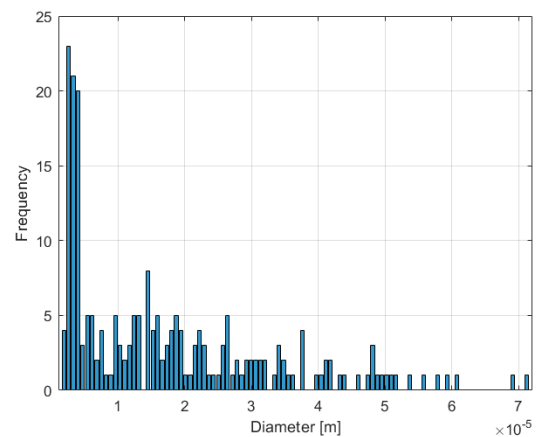


Figure 4: Particle size distribution for 1 [ms]

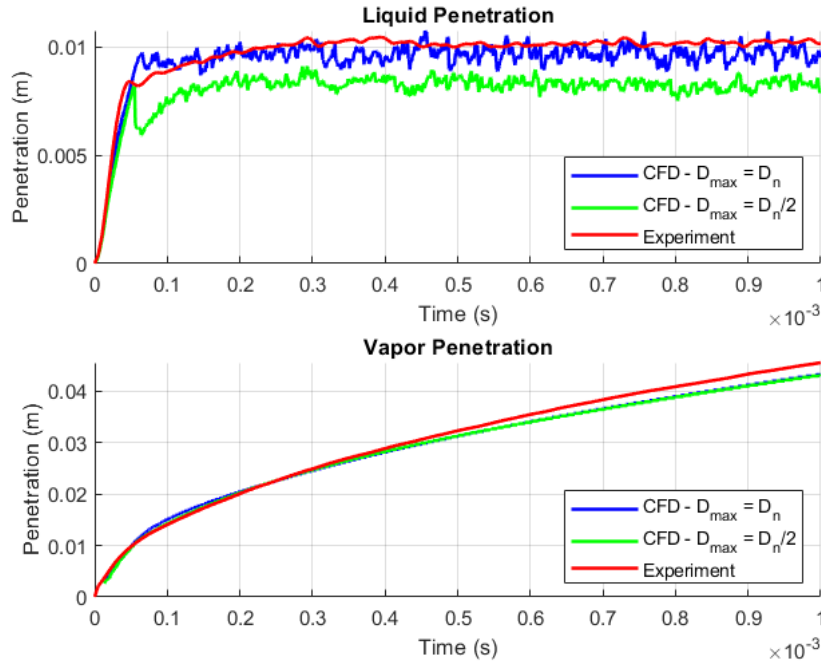


Figure 5: Simulation results showing liquid penetration and vapour penetration for a halved maximum particle diameter.

## 2.4 Influence of Rosin-Rammler distribution

The Rosin-Rammler distribution is a statistical distribution used to model the primary breakup of the particles after injection [4]. The probability distribution function  $p(D)$  and cumulative distribution  $P(D)$  are given by equations 1 and 2,

$$p(D) = \frac{q}{D_{\text{ref}}} \left( \frac{D}{D_{\text{ref}}} \right)^{q-1} \exp \left[ - \left( \frac{D}{D_{\text{ref}}} \right)^q \right] \quad (1)$$

$$P(D) = 1 - \exp \left[ - \left( \frac{x}{D_{\text{ref}}} \right)^q \right] \quad (2)$$

where  $D_{\text{ref}}$  is the reference diameter,  $D$  the particle diameter and  $q$  the Rosin-Rammler exponent.

The influence of the Rosin-Rammler distribution is studied by decreasing the maximum allowable particle diameter  $D_{\text{max}}$  by a factor of two. The results of this adjustment yield the results in Figure 5, which indicate no change in the vapour penetration, where the Pearson correlation coefficient remains 0.999. However, the liquid penetration has decreased, dropping the correlation to a value of 0.774.

Since the particle size has been decreased, the surface area is larger leading to a faster evaporation rate. This can be seen by the shorter liquid penetration length which ultimately leads to more optimal combustion.

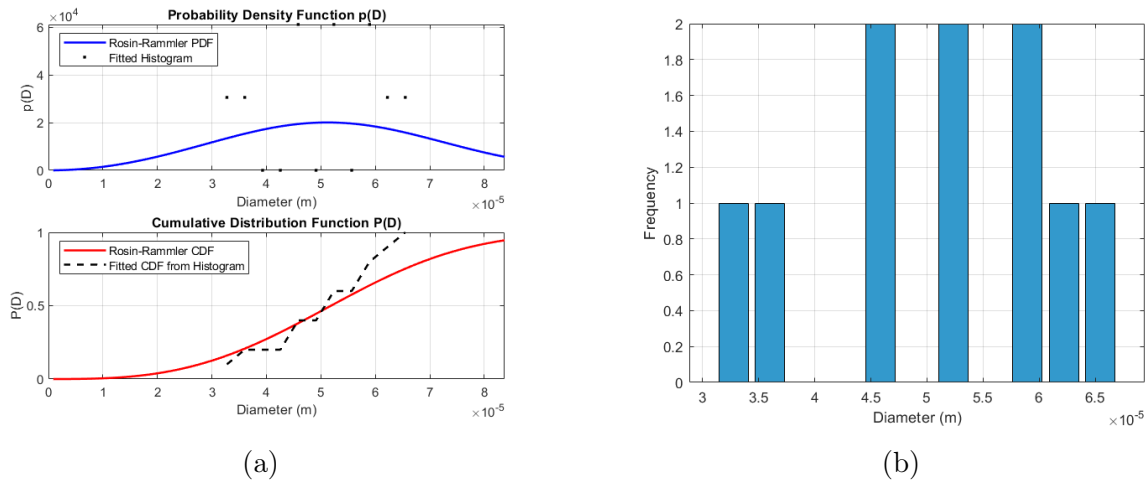


Figure 6: Plots showing the (a) probability and cumulative density function and (b) corresponding particle size distribution for 10 parcel streams at one timestep.

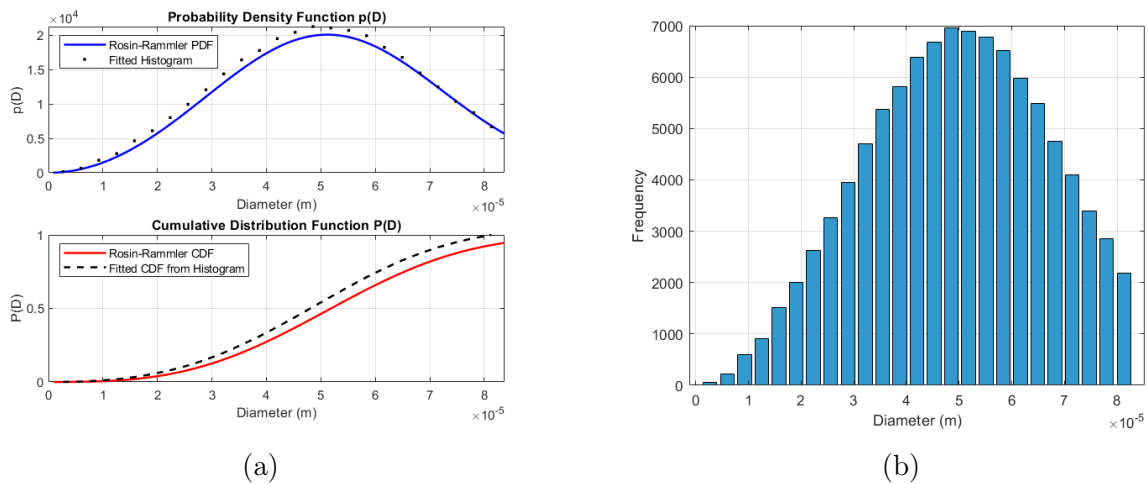


Figure 7: Plots showing the (a) probability and cumulative density function and (b) corresponding particle size distribution for 100,000 parcel streams at one timestep.

## 2.5 Influence of Parcel Streams

The number of parcel streams per injection point is the number of groups to which smaller particles in the flow are grouped. This quantity has no influence on the mass or volume flow rate from the injector, but it does influence the accuracy [5].

Consider Figure 6 showing how the parcel streams correspond to the Rosin-Rammler distribution. Based on Figure 6b it can be seen that the distribution faintly resembles Rosin-Rammler. The cumulative probability density function somewhat supports this, however, this number of parcel streams is not large enough to confidently state that the Rosin-Rammler distribution is obeyed.

On the other hand, 100,000 parcel streams show a very strong correlation between the particle size distribution at the injector and the Rosin-Rammler distribution, as shown in Figure 7. The probability density functions are shown in Figure 7a both indicating strong correlations between the two distributions. The Rosin-Rammler distribution can be seen more clearly in the shape of the histogram given in Figure 7b.

### 3 Part 2 - Combustion

Combustion modelling consists of simulating the chemical processes and physical phenomena that occur during combustion. These chemical kinetics can be modelled using different chemical mechanisms depending on the desired level of detail [6].

In this study, two different mechanisms, the reduced mechanism (Yao54) and the global mechanism are utilized to simulate the combustion process. The Yao54 mechanism is a detailed chemical kinetic mechanism that includes a set of 54 species  $N$  and 325 reactions  $M$  [7]. This model provides detailed kinetics however suffers from large computational costs. On the other hand, the global mechanism is a highly simplified model that has a small number of global reactions. This means the reaction is simulated in less detail but is less costly. Consider the temperature plots of both reactions in Figures 8 and 9.

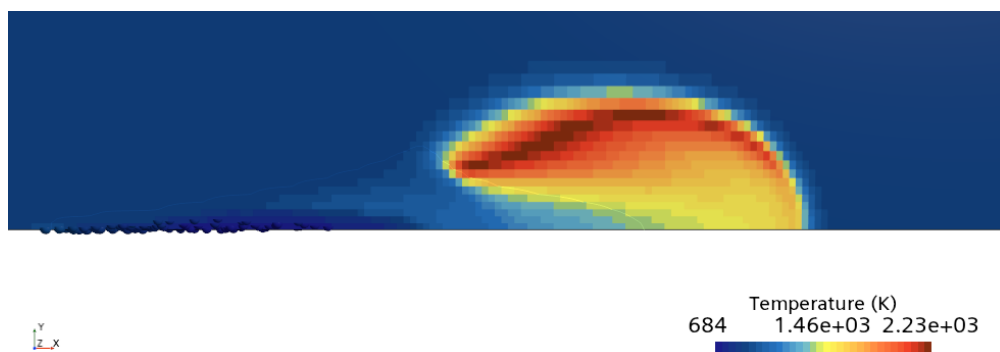


Figure 8: Yao54 mechanism temperature scalar scene

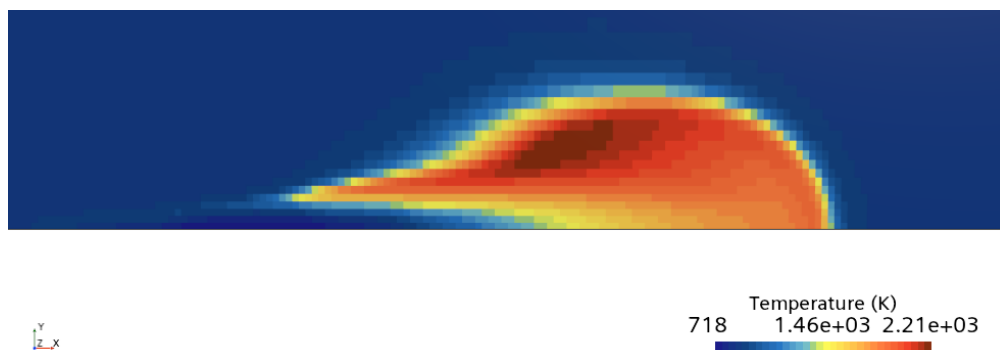


Figure 9: global mechanism scalar Scene

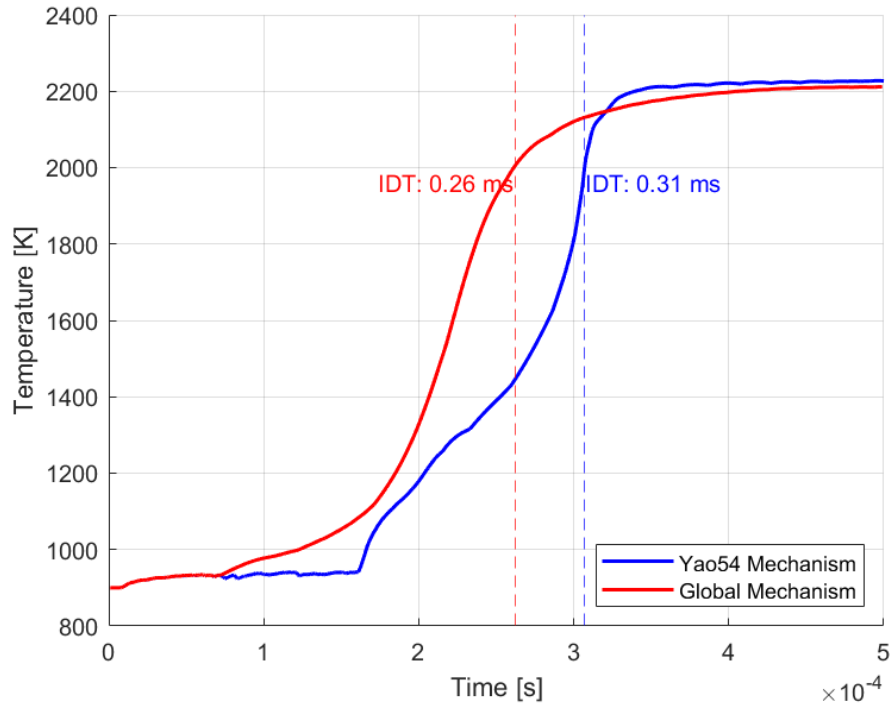


Figure 10: Maximum Temperature Plots from both mechanisms with their respective IDT

### 3.1 Model comparison

As illustrated in Figures 8 and 9, the experimental data from Yao54 demonstrates a maximum combustion temperature of 2,230 K and the simulation using global mechanisms recorded a slightly lower peak temperature of 2,210 K. A key difference between the two mechanisms is the time at which the temperature has begun to rise. Both methods appear to have identical temperature readings up to approximately  $0.8 \times 10^{-4}$  [s], where the global mechanism records a temperature rise. It is not until roughly  $1.7 \times 10^{-4}$  [s] that the Yao54 mechanism predicts the temperature rise. This difference is likely in the way that the mechanism sets the conditions for the reaction to occur. Another is the difference in the continuity of the curves. The global mechanism predicts a smoother curve presumably due to the more straightforward kinetic model.

### 3.2 Ignition delay timing

Figure 10 illustrates that both mechanisms resemble the characteristic maximum gas temperature profile for spray combustion. However, a noticeable difference can be highlighted in the Ignition delay time (IDT). The global mechanism simulation predicts an IDT of 0.26 [ms] whereas the Yao54 mechanism estimates 0.31 [ms]. This yields a difference of 0.05 [ms] between the two mechanisms.

IDT is a critical parameter in combustion, as it represents the period between fuel injection and the onset of combustion. A shorter IDT, as observed with the global mechanism

simulation, implies that the reactants reach their ignition conditions quicker, which can lead to an unexpected energy release and potentially affect the combustion stability. This earlier initiation may be attributed to the simplified chemical kinetics and higher reaction rates modelled in the global mechanism, which often fails to capture the detailed kinetics of the fuel oxidation process.

### 3.3 Choice of Mechanism

Considering the choice between the global mechanism, Yao54 and a more detailed model consisting of 100 species and 500 reactions, the application purpose must be taken into account.

#### 3.3.1 Wedge constant volume chamber

For the small-scale situation considered in this report, implementing a more detailed mechanism may be of interest for detailed studies. The 100 species and 500 reactions mechanism would provide more detailed results with more accurate kinetics such as intermediate species and stoichiometry [6]. Due to the larger number of species and reactions, the computational costs would increase as the number of algebraic and differential equations would be significantly larger. Assuming that the computational costs scale with  $\mathcal{O}(N \cdot M)$  these costs may be worthwhile for certain applications and studies.

#### 3.3.2 Full 3D diesel engine

For a full 3D diesel engine including piston motion and valves, the global mechanism is the wiser choice of mechanism. In such a setup the amount of detail is less important and computational costs are too high. Even if the simulation were completed, it may not necessarily yield more accurate results that capture engine dynamics. The global mechanism can be considered sufficient for capturing the essential combustion characteristics needed in a full 3D diesel engine simulation.

## 4 Conclusion

This report showcased the general approach to modelling combustion in a wedge constant volume chamber using lagrangian spray modelling. The findings indicate the expected liquid and vapour penetration lengths along with their discrepancies and preferable simulation settings. The comparison between different mechanisms aimed at capturing the chemical kinetic highlighted the balance between accuracy and computational costs depending on the application.

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## Nomenclature

CFD Computational fluid dynamics

IDT Ignition Delay Time

## References

- [1] “Diesel Data Search 2024 national technology and engineering solutions of sandia.”
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